

Section A – Multiple Choice Questions (10 × 1 = 10 Marks)

1. Which statement is used to skip the rest of the code inside a loop for the current iteration?

- a) break b) continue c) pass d) exit

2. Which loop is guaranteed to run at least once in Python?

- a) for b) while c) do-while d) None

3. What will be the output of:

```
for i in range(5):
```

```
    if i == 3:
```

```
        break
```

```
    print(i)
```

- a) 0 1 2 3 b) 0 1 2 c) 3 4 d) 0 1

4. The pass statement in Python:

- a) Stops the loop immediately
b) Skips one iteration of the loop
c) Does nothing and acts as a placeholder
d) Raises an error

5. Which of the following is not a conditional statement?

- a) if b) elif c) else d) switch

6. What is the output of:

```
count = 0
```

```
while count < 3:
```

```
    count += 1
```

```
    print(count)
```

- a) 0 1 2 b) 1 2 3 c) 0 1 2 3 d) Infinite loop

7. In `for i in range(3)`, the variable `i` takes values:

- a) 0 1 2 b) 1 2 3 c) 0 1 2 3 d) None

8. Which of the following can be used as an infinite loop?

- a) `while True:` b) `for i in range(10):` c) `while False:` d) None

9. In a nested loop, how do you break out of only the inner loop?

- a) Use `exit()` b) Use `break` inside inner loop c) Use `pass` d) Impossible

10. What is the correct syntax of if statement in Python?

- a) `if condition:` b) `if (condition)` c) `if {condition}` d) `if condition`

Section B – Short Answer Questions (5 × 2 = 10 Marks)

1. Explain the difference between `break` and `continue` with examples.

2. Write a while loop to print numbers from 1 to 5.

3. What is the purpose of pass in Python? Give one example.

4. Write the output of the following code:

```
for i in range(4):  
    if i == 2:  
        continue  
    print(i)
```

5. Mention two differences between for loop and while loop.

Section C – Descriptive Questions (6 × 5 = 30 Marks)

1. Explain all conditional statements in Python with syntax and example code.

2. Describe the working of a for loop in Python. Write a program to print the multiplication table of a given number.

3. What are transfer statements? Explain break, continue, and pass with examples.

4. Write a program using while loop to sum all even numbers between 1 and 20.

5. Write a Python program that asks the user for a number and checks if it is positive, negative, or zero using if-elif-else.

6. Write a program using nested loops to print the following pattern:

```
*  
* *  
* * *  
* * * *
```